Fiscal Year:	FY 2011	**Task Last Updated:**	FY 12/20/2011
PI Name:	Keeton, Kathryn Ph.D.		
Project Title:	Analog Validation Study of Team Measures		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	kathryn.keeton@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	281-204-1519
Organization Name:	Wyle/NASA Johnson Space Center		
PI Address 1:	1290 Hercules Drive, Suite 120		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
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No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Shea, Cami1e	**Contact Phone:**	281-244-2017
Contact Email:	shea@dsls.usra.edu		
Flight Program:			
Flight Assignment:			
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COI Name (Institution):	Leveton, Lauren (NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
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Performance Goal Text:			
	The NASA/JSC Behavioral Health and Performance Element of Space Medicine Division, Human Research Program (BHP), is proposing a research study that will help identify the most optimal measures and monitoring tools to assess three important outcomes: crew performance, crew cohesion, and crew-ground interaction. The proposed study provides an opportunity to assess these important team outcomes and identify the most serious and important risk factors that impact these three factors, utilizing high fidelity analog environments. This data collection process will also contribute towards the development of a normative database that can be used to validate these measures for spaceflight. When considering long duration missions, it is acknowledged that there are many factors likely to influence crew performance, crew cohesion, and the interactions between crew and ground control. These factors have the potential to create interpersonal conflict, decrease morale, and/or create a negative environment for both individual crewmembers and the team itself. Many of these factors, if not addressed, may negatively influence cohesion among the crewmembers, interactions between the crew and ground control, and crew performance. In order to mitigate these		

Task Description:	negative consequences, it is essential to conduct research that will identify the most optimal measures and tools to monitor and assess these risk factors and their impact on these important team outcomes. As described above, the main objective of this study is to assess crew performance, crew cohesion, and crew-ground interaction and influential risk factors in the context of high fidelity analogs in order to develop a normative database of team measures that may be generalizable to spaceflight and the astronaut population. However, the results of this study will also provide a standard set of team measures that will be used across analog research with BHP's Team Risk. In addition, the results of this study will enrich the team literature and evidence base of team research within the analog environment. This information will inform the implementation of the team research strategy in spaceflight. Thus a second objective of this study is to determine the feasibility of a standard set of team measures for spaceflight. The main objective of this study is to create a standardized set of measures for team gaps and a normative database of team measures that are generalizable to both the astronaut population and the spaceflight context by utilizing high fidelity analog environments. This study will provide the opportunity to assess crew performance, crew cohesion, and crew-ground interaction and identify which risk factors are most influential in relation to those three outcomes. In addition, it will also enrich the team literature and knowledge base of team research within the analog environment. Specific aims of the study include: (1) Selection of initial team measures-- measures will be selected based on present knowledge of important factors that relate to crew cohesion, crew performance, and crew-ground interaction as well as a consideration of acceptability and feasibility of implementing those measures selected (2) Data collection at high fidelity analogs to create a team measure normative database-- data will be collected at multiple analogs over time to create enough power to complete analyses (at least 25-30 teams) (3) Analyze collected data to determine optimal set of team measures-- will assess measures used and the robustness of the data (4) Conduct an annual review to update measures used and/or analogs to be utilized
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	This project is now complete; from our analyses, it is evident that additional data collection is needed in order to draw accurate and informative conclusions about behavioral health and performance issues experienced in isolated, confined, and extreme environments that are analogous to spaceflight.
Bibliography Type:	Description: (Last Updated:)